



Fostering intergenerational understanding: roles of intrapersonal and interpersonal mindfulness in communication

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Abstract

Population aging and declining birth rates have underscored the growing importance of intergenerational communication. This study employed a matched pairs experimental design to explore how intrapersonal (present-focused, non-judgmental attitude) and interpersonal (awareness of verbal and nonverbal cues in communication) mindfulness differently impact intergenerational communication quality. Forty-four Taiwanese intergenerational dyads engaged in three communication scenarios, and their communication satisfaction was assessed from emotional, cognitive, and behavioral perspectives based on the Interpersonal Chemistry Model. Results from the multilevel regression model showed that both intrapersonal and interpersonal mindfulness increased positive emotion and reduced stress (emotion), fostered more empathy, led to higher levels of communication satisfaction, facilitated closer psychological distance (cognition), and helped to reach a higher consensus on controversial topics (behavior). Compared to intrapersonal mindfulness, interpersonal mindfulness promoted more aspects of communication, including reducing negative emotion and accelerating cooperation speed. Results based on the Actor-Partner Interdependence Model indicated that both intrapersonal and interpersonal mindfulness had a broader enhancing effect on communication quality for older adults compared to younger adults. This study highlights the significant role of mindfulness in intergenerational communication and suggests that the design of future communication tools should incorporate elements of mindfulness.

Keywords Actor-partner interdependence model · Age difference · Communication quality · Intergenerational communication · Mindfulness

Introduction

Challenges of intergenerational communication

The worldwide increase in the population of older adults, coupled with a global decline in birth rates, has spurred the need for various forms of intergenerational communication, such as those involving long-term care facilities and intergenerational communities (WHO, 2022; Zhong et al., 2020; Turkle, 2011). However, Inter-generational communication often involves conflicts and challenges. For example, In the workplace, young people may face stereotypes from older colleagues due to their lack of social experience, which can affect recruitment, promotion, or salary increases (Wood et al., 2008). Furthermore, in parent-child communication, parents often struggle to recognize the situational culture of the new era and find it challenging to integrate the values of different generations. Similarly, children may find it difficult

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to understand the directions their parents hope for them to develop (Fong, 2007).

Ryan et al. (1995) introduced the Communication Predicament theory to elucidate the challenges in intergenerational communication. This theory suggests that when interacting with younger adults, older adults become particularly sensitive to negative age-related stereotypes and may over-adjust their behavior. This overadjustment can lead to significant communication stress. Additionally, in intergenerational family communication, younger individuals often perceive more criticism and rejection from older adults, which leads to greater communication stress (Zhang & Lin, 2009).

Intrapersonal mindfulness and interpersonal mindfulness

Mindfulness originated over 2,500 years ago in ancient North India and was initially taught by Siddhartha Gautama, who became known as “the Buddha,” meaning “the awakened one.” Siddhartha Gautama believed that individuals could train their minds to bring scattered thoughts back into focus, attend to present experiences, and concentrate on achieving their goals (Kang & Whittingham, 2010). In psychology and related fields, intrapersonal mindfulness refers to maintaining complete awareness of the present moment and adopting a curious, non-judgmental, and accepting attitude toward current experiences (Bishop et al., 2004; Brown & Ryan, 2003).

Baer et al. (2006) proposed that intrapersonal mindfulness can be treated as an individual’s trait variable and assessed across five facets: (a) Nonreactivity to inner experience, (b) Observing thoughts and feelings, (c) Acting with awareness, (d) Describing with words, and (e) Nonjudging of inner thought. Meta-analysis studies have found that individuals with higher levels of intrapersonal mindfulness also demonstrate higher emotional intelligence in life, which means they possess better abilities to perceive, understand, and manage their own emotions (Miao et al., 2018; Mesmer-Magnus et al., 2017).

However, Pratscher et al. (2018) highlighted that intrapersonal mindfulness predominantly emphasized awareness of one’s internal psychological and bodily sensations, with less focus on extending this mindfulness to the level of interpersonal relationships. Therefore, they developed the concept of interpersonal mindfulness to gauge whether individuals maintained a receptive awareness of what was occurring throughout interactions, including their own and others’ thoughts, emotions, feelings, bodily sensations, experiences, and intentions.

Enhancing effects of intrapersonal and interpersonal mindfulness in the intergenerational communication process

Both intrapersonal mindfulness and interpersonal mindfulness have the potential to enhance communication quality. Research has found that specific dimensions of intrapersonal mindfulness, such as nonjudging and describing, were associated with precisely understanding linguistic and nonverbal cues during communication (Jones & Hansen, 2015, study 1). Furthermore, compared to control groups, individuals who have undergone intrapersonal mindfulness training demonstrate better communication and social skills (Jones & Hansen, 2015, studies 2 and 3). Higher intrapersonal mindfulness individuals were also associated with higher communication satisfaction levels and experiencing less stress during communication when discussing conflict topics with committed partners (Barnes et al., 2007; Seidel et al., 2020). Mindfulness can also help resolve the challenges of intergenerational communication between parents and children: Highly mindful parents perceived lower stress in the parenting process (Gouveia et al., 2016), more positive parenting practices, and decreased children’s problem behaviors (Han et al., 2021). Based on the above evidence, we believe that merely increasing mindfulness awareness from the self can have positive effects on communication quality.

When mindfulness awareness expands from the self (intrapersonal mindfulness) to include both parties in the communication (interpersonal mindfulness), its positive impact on intergenerational communication may become even stronger. Sofer (2018) suggested that interpersonal mindfulness training systems can effectively enhance mutual understanding and reduce the occurrence of violent verbal communication, such as belittling, blaming, harsh judgment, criticism, and derogative comparisons. Previous research also found that, under high levels of perceived stress, teachers high on interpersonal mindfulness were substantially more emotionally supportive toward students in the classroom than teachers low on interpersonal mindfulness. Interpersonal mindfulness hereby may buffer against the negative effect of stress on teachers (Molloy Elreda et al., 2019). Prince-Paul and Kelley (2017) also argued that interpersonal mindfulness-based communication can improve caregiver-patient relationships, thereby promoting patient-centered healthcare systems. Duncan et al. (2009) suggest that interpersonal mindfulness can be applied in the parenting process, cultivating higher parenting and communication skills in parents, which in turn can enhance children’s well-being and self-regulation abilities. In addition, Pratscher et al. (2019) found that interpersonal mindfulness had a stronger predictive power for friendship quality

compared to intrapersonal mindfulness. This suggests that interpersonal mindfulness is likely a key indicator for capturing high-quality communication.

This study hypothesizes that interpersonal and intrapersonal mindfulness can also address communication difficulties in intergenerational interactions, thereby enhancing the quality of intergenerational communication. Furthermore, while past research has independently explored the enhancing effects of interpersonal mindfulness (Jones & Hansen, 2015; Barnes et al., 2007; Seidel et al., 2020) and intrapersonal mindfulness (Molloy Elreda et al., 2019; Prince-Paul & Kelley, 2017; Duncan et al., 2009), there has been no comparison between the two kinds of mindfulness' effects. We hypothesize that interpersonal mindfulness, which emphasizes greater interpersonal awareness, has a stronger positive impact on intergenerational communication compared to intrapersonal mindfulness.

Age-related differences in the enhancing effects of mindfulness within an intergenerational communication context

Past research has found that older adults exhibit higher levels of intrapersonal mindfulness than younger adults, which results in older adults experiencing more positive and less negative emotions in daily life (Raes et al., 2015; Shook et al., 2017). Additionally, studies have shown that certain intrapersonal mindfulness elements, such as present-moment attention and nonjudgment, have an increasingly strong relationship with well-being as age increases (Mahlo & Windsor, 2021). Furthermore, intrapersonal mindfulness has been found to have a greater positive impact on relationship satisfaction among older adults than younger adults (Lenger et al., 2019).

Although the above-mentioned studies indicate that the enhancing effects of intrapersonal mindfulness are greater for older adults than for younger adults, past studies have primarily observed the impact of mindfulness on younger and older adults separately, without analyzing the age-related difference of mindfulness's enhancing effect within the intergenerational communication context. The above-mentioned studies have also focused on exploring age differences in intrapersonal mindfulness without investigating age differences in interpersonal mindfulness. Therefore, this study aims to explore whether, within the context of intergenerational communication, the enhancing aspects of intrapersonal and interpersonal mindfulness are more extensive for older adults than younger adults.

The present study

This study seeks to distinguish between intrapersonal and interpersonal mindfulness, exploring how each type enhances the quality of intergenerational communication and comparing their impacts on its various aspects. Furthermore, the study aspires to examine any generational differences in the impact of mindfulness on communication and whether the mindfulness of one partner affects the communication quality of the other partner. Drawing from the Interpersonal Chemistry Model proposed by Reis et al. (2021), the study intends to capture the panorama of communication quality from three perspectives: Affective, Cognitive, and Behavioral.

In the Interpersonal Chemistry Model (see Fig. 1), the Affective perspective refers to “affective positivity to others,” which means both parties share a positive affective component during their communication process. Previous literature indicates that positive and negative emotions, as well as perceived stress, determine the quality of intergenerational communication (Bakalim & Taşdelen-Karçay, 2015; Jiménez et al., 2019). Therefore, we intended to capture this dimension by measuring both positive and negative emotion, as well as stress levels, to fully understand these emotional dynamics.

Cognitive perspective refers to the “perception of shared identity,” which means both parties share the same identity during their interaction process. In intergenerational communication, participants' communication satisfaction has a positive relationship with the degree of shared identity in communication (Soliz et al., 2009). Psychological distance and state empathy are also key psychological variables for expressing mutual comprehension in communication (Gallesse, 2005; Boothby et al., 2016). Therefore, we planned to capture this dimension using measures of communication satisfaction, state empathy, and perceived psychological distance between individuals to thoroughly assess the nuances of relational perception.

The behavioral perspective refers to “coordinated goal-relevant activity”, which means both parties effectively coordinate and collaborate to achieve designated goals during their interaction process. Previous research suggests that in the process of intergenerational communication, conflict management is inseparable from high levels of cooperation between generations (Sorenson, 1999). In addition, a study also shows that communication effectiveness is the foundation of good intergenerational relationships (Bearison & Casel, 1975). We designed discussions on controversial issues and a tangram task to evaluate and strategize the participants' performance during the test, aiming to measure how effectively the participants collaborate and communicate in a controlled setting. Additionally, we believe that within the

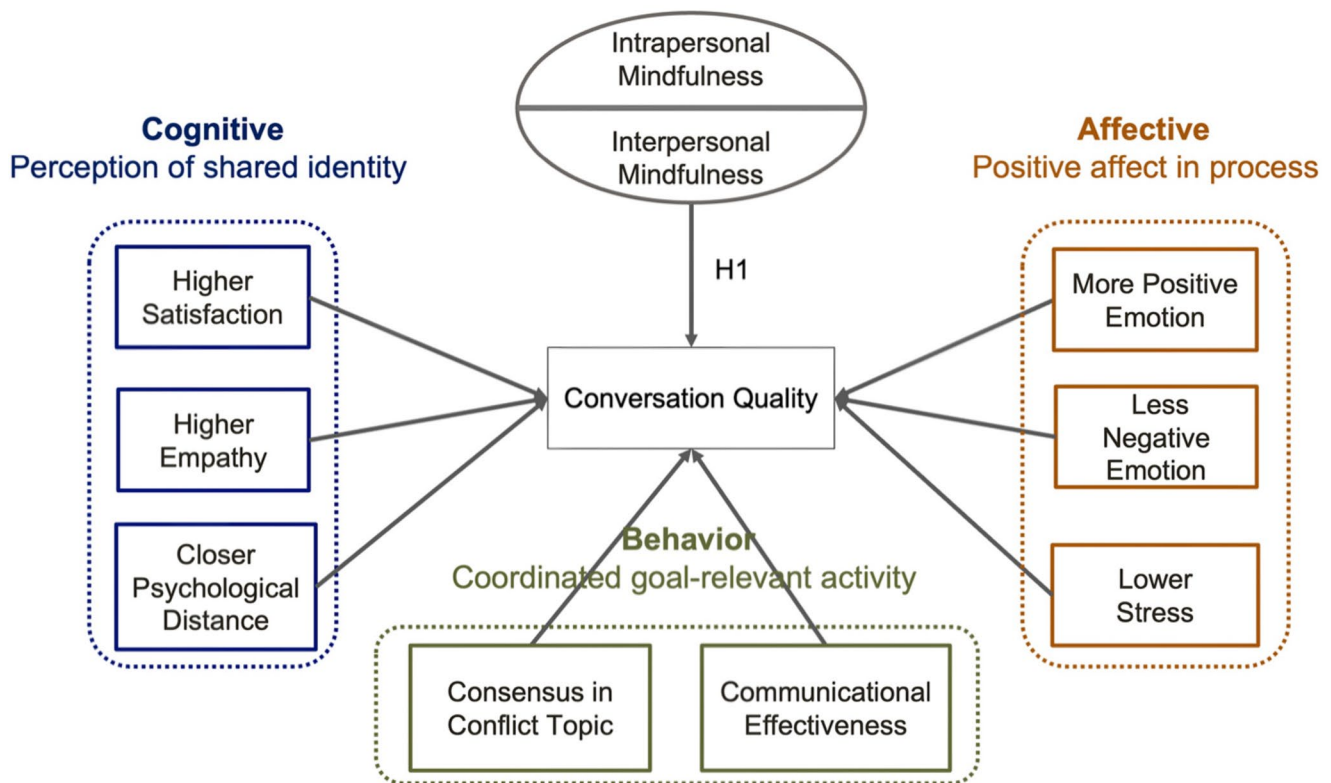


Fig. 1 Framework of exploring the effect of intrapersonal and interpersonal mindfulness on communication quality based on the Interpersonal Chemistry Model (Reis et al., 2021)

context of intergenerational communication, interpersonal mindfulness can enhance more aspects of communication quality compared to intrapersonal mindfulness.

Hypotheses:

- Hypothesis 1 interpersonal mindfulness and intrapersonal mindfulness both promote intergenerational communication quality.

– *Affective: Affective Positivity to Others*

H1-a: Interpersonal mindfulness and intrapersonal mindfulness increase one's perception of positive emotion during intergenerational communication.

H1-b: Interpersonal mindfulness and intrapersonal mindfulness decrease one's perception of negative emotion during intergenerational communication.

H1-c: Interpersonal mindfulness and intrapersonal mindfulness reduce one's stress during intergenerational communication.

– *Cognitive: Perception of shared identity*

H1-d: Interpersonal mindfulness and intrapersonal mindfulness enhance one's perceived satisfaction during intergenerational communication.

H1-e: Interpersonal mindfulness and intrapersonal mindfulness encourage the use of more empathy during intergenerational communication.

H1-f: Interpersonal mindfulness and intrapersonal mindfulness promote a closer psychological distance between individuals during intergenerational communication.

– *Behavioral: Coordinated goal-relevant activity*

H1-g: Interpersonal mindfulness and intrapersonal mindfulness lead to a higher perceived consensus on controversial issues after discussion during intergenerational communication.

H1-h: Interpersonal mindfulness and intrapersonal mindfulness facilitate faster completion of cooperative tasks between individuals during intergenerational communication.

- Hypothesis 2 Interpersonal mindfulness enhances more aspects of intergenerational communication quality compared to intrapersonal mindfulness.

- Hypothesis 3 The enhancing aspects of interpersonal and intrapersonal mindfulness are more extensive for older adults than younger adults during intergenerational communication.

Methods

Participants

The participants comprised 44 younger adults (Mean Age = 21.95, SD = 3.13, 23 female) and 44 older adults (Mean Age = 62.61, SD = 6.28, 22 female), all of whom were Taiwanese, spoke fluent Mandarin and were recruited through online platforms or advertising flyers. The majority (69.3%) held a Bachelor's degree, and an additional 26% earned a Master's degree. All participants provided written informed consent and agreed to data publication terms before study participation. The experimental procedures and materials received approval from the ethical review board (NTU-REC-202007HS058) prior to the commencement of the study.

Each participant was invited to the university laboratory in Taipei to engage in communication dialogues and received a compensation of 400 New Taiwan Dollars after the conversation. Only participants who could communicate fluently in Chinese were recruited. After the communication sessions, researchers reviewed the recorded videos to determine if any language communication barriers existed between the interlocutors. Upon review, all participants were found to be capable of fluent intergenerational communication, and therefore all participants' data were included in the analysis.

Study design and instruments

This study employs a 2×2 matched-pairs design, with intrapersonal mindfulness differences as the between-groups variable and age differences as the within-groups variable. Before participants took part in the study, we collected their demographic information and intrapersonal mindfulness scores through an online questionnaire. The intrapersonal mindfulness scores were calculated based on their responses to the 39-item Five-Facet Mindfulness Questionnaire (FFMQ), which was developed around five distinct factors and has been subsequently validated (Baer et al., 2006). An example item is "When I'm walking, I deliberately notice the sensations of my body moving."

Subsequently, participants were classified according to their intrapersonal mindfulness scores, with a cutoff of 135 (Lilja et al., 2013), whereby a score above 135 was

indicative of high mindfulness, and a score below 135 was indicative of low mindfulness. Following the aforementioned principle, participants were divided into one of three groups: Experiment, Mixed, and Control. In each group, one younger adult and one older adult were paired up as a communication dyad if they held quite contrasting opinions on at least two topics. Intergenerational dyads in the Experiment group consisted of two participants exhibiting high mindfulness, the Mixed group consisted of one participant with high mindfulness and one with low mindfulness, and the Control group consisted of two participants with low mindfulness. Also appraised were their views on eight distinct topics - career choices, dietary beliefs, romantic partner selection, education, climate change, renewable energy, wealth inequality and income redistribution, and financial management. Due to practical considerations, we aimed to exclude confounding variables that significantly affect communication quality in previous research, such as differing family roles and the varying durations of family conflicts (Cooke & Rousseau, 1984; Roubinov & Luecken, 2013). Therefore, we opted for a more controlled approach by pairing strangers in an intergenerational communication dyad.

Upon participants' arrival at the laboratory and prior to the commencement of the formal intergenerational communication, two questionnaires were administered. One was the Positive and Negative Affect Scale (Watson et al., 1988), used to assess the participants' emotional states with sample items including "Interested" for positive affect and "Distressed" for negative affect. The other questionnaire was the psychological distance scale adapted from the Inclusion of Other in the Self scale (Aron et al., 1992), where seven pairs of circles were presented with a number (1–7) beneath each pair. One circle in each pair labeled "self" represented the participant himself/herself, and the other circle labeled "other" represented his/her assigned partner. Participants were instructed to choose a pair of circles that best described the relationship with their partners at that moment by selecting the corresponding number. The higher the number, the closer they felt toward their partners.

Following the conclusion of the intergenerational communication, participants were required to complete several post-test questionnaires. The psychological distance questionnaire and Positive and Negative Affect Scale were administered again during the post-test assessment. Communication satisfaction was measured using the scale developed by Goodboy et al. (2009), with a sample item being "My conversations with my partner are worthwhile." Perceived empathy was assessed using Shen's (2010) scale, with a sample item being "The character's emotions are genuine." Communication stress was measured with the scale by Antony et al. (1998), with a sample item being "I felt I was rather touchy." Finally, interpersonal mindfulness

was evaluated using Pratscher et al.'s (2019) scale, with a sample item being "When I am conversing with another person, I am fully engaged in the conversation. All reliability indicators for the relevant questionnaires are presented in Table 1. In addition, all questionnaires and materials used in this study are included in Supplemental Materials A.

Procedure

The entire intergenerational communication session lasted approximately one hour. Following a brief 5-minute self-introduction (middle panel, Fig. 2), the main experiment, consisting of three sessions, would officially start.

In the first session, with each intergenerational dyad, the participants would take turns to share a positive and a negative experience, purporting to gauge their levels of empathy and focusing on the present. The listener was encouraged to offer timely feedback or to interrupt for clarification instead of merely listening. Each story was slotted for 5 min, so the session lasted 20 min. Participants were allowed to decide between them the sequence of sharing. Once ready, they would start an automatic timer nearby, which would beep every 5 min to remind them to move to the next experience sharing.

In the second session, each dyad was asked to complete eight tangram games as fast as possible. Tangram is a puzzle game consisting of seven flat polygons assembled to replicate a pattern using all pieces without overlap. Four of the eight tangram games were selected from Marshall and Novick's (1995) tasks (set A). The selection criteria for the tangram models were based on a pilot study, which identified models with an appropriate level of difficulty. All participants in the pilot study were able to complete these models within five minutes. Another four were chosen from the Internet (set B) because the totally six patterns from Marshall and Novick's (1995) tasks were not enough to allow equal rotations for the two participants to act as instructors and constructors, respectively. Each participant was given either set A or set B, with each pattern printed on a piece of paper held in a file binder. With every two patterns, they took turns as an instructor or a constructor. The instructor would verbally instruct the constructor to build a pattern in the instructor's file binder, which the constructor could not see. The constructor would follow the instructions to arrange the pieces into the pattern. The instructor was not allowed to rely on any nonverbal means, such as gestures or directly moving pieces. The time spent to finish each pattern correctly was recorded, serving as a measurement of how effective their verbal communication and collaboration were. They were also informed that if it took more than 10 min, they would be recommended to move on to the next pattern, and the current pattern was recorded as a failure.

Table 1 Descriptive statistics, reliability and correlation

Variable	<i>M</i>	<i>SD</i>	Cronbach's alpha	Correlation									
				1	2	3	4	5	6	7	8	9	10
1. Intrapersonal mindfulness	132.39	19.20	0.73	1									
2. Interpersonal mindfulness	96.39	11.53	0.81	0.37***	1								
3. Post_positive emotion	35.70	6.64	0.87	0.56***	0.56***	1							
4. Post_negative emotion	18.19	6.76	0.92	-0.26*	-0.17	-0.11	1						
5. Stress	8.16	3.76	0.76	-0.46***	-0.34***	-0.40***	0.35***	1					
6. Empathy	62.98	7.40	0.94	0.27*	0.68***	0.47***	-0.25*	-0.17	1				
7. Communication satisfaction	26.76	3.12	0.86	0.27**	0.61***	0.46***	-0.18	-0.13	0.76***	1			
8. Closer psychological distance	4.28	1.52	-	0.31**	0.29***	0.38***	-0.15	-0.20	0.26*	0.45***	1		
9. Consensus in conflict topic	5.08	0.91	-	0.22*	0.44***	0.41***	-0.11	-0.05	0.59***	0.50***	0.40***	1	
10. Time finishing tangram	584.77	172.39	-	-0.10	-0.23*	0.00	0.01	-0.02	-0.19	-0.08	-0.08	0.00	1

The table only lists the post-test values of positive and negative emotion without including pre-test values, since they are used for statistical control and are not considered primary explanatory outcomes. "Time finishing in tangram" refers to the total time (Second) participants took to complete all designated tangram tasks. Closer psychological distance, Consensus in conflict topic, and Time finishing tangram were measured using only one item or indicator, therefore reliability results were not indicated. *N* = 88. **p* < .05; ***p* < .01; ****p* < .001

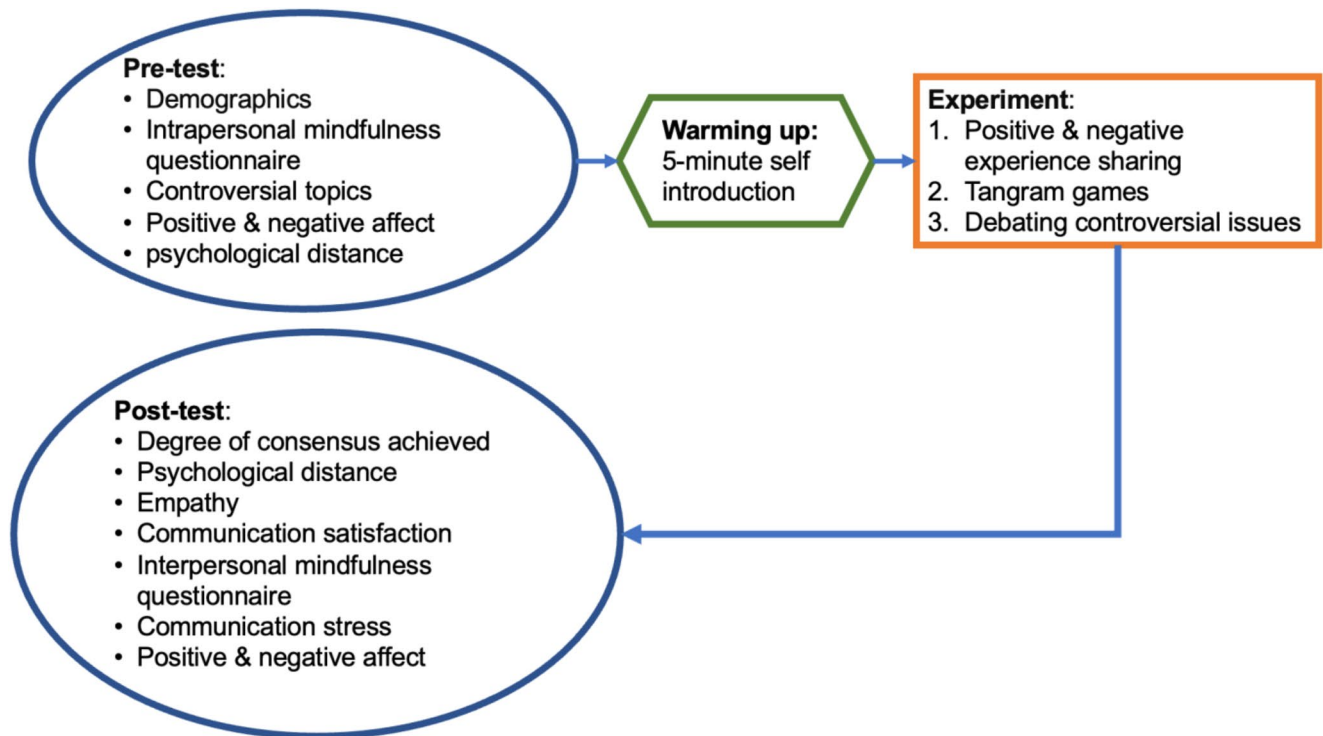


Fig. 2 The flow of the experiment. The arrangement of the questionnaires in the diagram from top to bottom represents the order in which participants filled them out

In the third session, two controversial issues, which the participants in each dyad mostly disagreed upon based on their views on the aforementioned various topics, were brought up for them to debate, aiming to appraise how they resolved disagreements and try to reach a consensus. It was stressed that they should find the point(s) of dispute between them, clarify it/them, and reach an agreement in the end. This session lasted 20 min. The full instructions for carrying out the experiment are included in Supplemental Materials B.

Data analysis

Data were analyzed using R (Version 4.1.2) and RStudio (2023.06.2). A preliminary analysis revealed normally distributed intrapersonal mindfulness scores ($M = 132.39$, Range = 79–188, $SD = 19.2$). The Shapiro-Wilk test confirmed normal distribution ($W = 0.99$, $p = .89$, $n = 88$), and quantile-quantile plots showed close alignment with the theoretical normal line (See Supplemental Materials C).

As intrapersonal mindfulness scores demonstrated a normal distribution, categorizing participants based solely on discrete class variables while disregarding the continuous nature of the data was deemed methodologically inappropriate. A multilevel regression model was employed to analyze the data and address this issue. The “lme4” package in R (version 1.1.29) was used to set up a multilevel

regression model and examine the influence of intrapersonal and interpersonal mindfulness on communication quality. Both intrapersonal and interpersonal mindfulness scores were treated as fixed effects in the model while controlling for the variance of intercepts in the random effects. In addition, to detect changes in affect and cognition, we controlled for pretest scores in positive and negative emotion, as well as psychological distance, when using mindfulness to predict posttest emotion and psychological distance. We also employed the “name” package in R (version 3.1.157) to conduct an Actor-Partner Interdependence Model (APIM) analysis to explore differences in cross-generational communication performance between older and younger adults (Kenny et al., 2020).

Results

There was a moderate positive correlation between the scores of interpersonal and intrapersonal mindfulness ($r = .37$, $p < .001$), both of which positively correlated with the majority of dependent variables measured in this study (see Table 1).

Impact of intrapersonal and interpersonal mindfulness on communication quality

This study first examined whether intrapersonal and interpersonal mindfulness would influence individuals' perception of communication quality (see Table 2). Firstly, at the affective level, it was found that individuals with higher levels of interpersonal and intrapersonal mindfulness reported significantly increased positive emotion (intrapersonal mindfulness: $\beta = 0.076$, $SE = 0.024$, $t = 3.129$, $p = .002$; interpersonal: $\beta = 0.152$, $SE = 0.037$, $t = 4.044$, $p < .001$) and less communication stress (intrapersonal: $\beta = -0.09$, $SE = 0.019$, $t = -4.867$, $p < .001$; interpersonal: $\beta = -0.110$, $SE = 0.033$, $t = -3.320$, $p = .001$). Secondly, at the cognitive level, individuals with higher levels of interpersonal and intrapersonal mindfulness exhibited higher communication satisfaction (intrapersonal: $\beta = 0.042$, $SE = 0.017$, $t = 2.539$, $p = .013$; interpersonal: $\beta = 0.162$, $SE = 0.023$, $t = 6.939$, $p < .001$) and higher level of empathy (intrapersonal: $\beta = 0.102$, $SE = 0.040$, $t = 2.536$, $p = .013$; interpersonal: $\beta = 0.441$, $SE = 0.050$, $t = 8.669$, $p < .001$). They reported feeling closer psychological distance with their communication partners after the interaction (intrapersonal: $\beta = 0.018$, $SE = 0.008$, $t = 2.285$, $p = .002$; interpersonal: $\beta = 0.035$, $SE = 0.012$, $t = 2.766$, $p = .006$). Lastly, at the behavioral level, individuals with higher interpersonal and intrapersonal mindfulness

perceived reaching higher degree of consensus with their partners after discussing controversial topics (intrapersonal: $\beta = 0.010$, $SE = 0.005$, $t = 2.067$, $p = .042$; interpersonal: $\beta = 0.034$, $SE = 0.007$, $t = 4.536$, $p < .001$).

Meanwhile, only the participants with high interpersonal mindfulness managed to reduce their negative emotion during the communication process ($\beta = -0.092$, $SE = 0.046$, $t = -2.020$, $p = .047$) and complete the tangram puzzle task faster ($\beta = -3.328$, $SE = 1.580$, $t = -2.106$, $p = .038$). On the contrary, intrapersonal mindfulness had no significant impact on the participants' negative emotion ($\beta = -0.039$, $SE = 0.028$, $t = -1.397$, $p = .166$) and the speed of completing tangram puzzle task ($\beta = -0.908$, $SE = 0.967$, $t = -0.939$, $p = .350$) during the communication process.

Considering the influence of mindfulness on various aspects of communication, this study utilized the APIM (Table 3) to control for intergenerational effects and examine how the intrapersonal mindfulness scores of younger and older participants affect their own perceived communication quality, known as *the actor effect*.

The results showed that higher intrapersonal mindfulness in younger adults was associated with experiencing less stress during communication ($\beta = -0.095$, $SE = 0.030$, $t = -3.182$, $p = .002$) while higher intrapersonal mindfulness in older adults was associated with higher positive emotion ($\beta = 0.075$, $SE = 0.036$, $t = 2.067$, $p = .042$), higher

Table 2 Summary of the multi-level influence of mindfulness

	Estimated Coefficient (β)	SE	t	p
Intrapersonal mindfulness predicting dependent variables				
Affective aspect				
Post positive emotion	0.076	0.024	3.129	0.002**
Post negative emotion	-0.039	0.028	-1.397	0.166
Post stress	-0.090	0.019	-4.867	< 0.001***
Cognitive aspect				
Communication satisfaction	0.042	0.017	2.539	0.013*
Post empathy	0.102	0.040	2.536	0.013*
Post psychological distance	0.018	0.008	2.285	0.002**
Behavioral aspect				
Consensus in conflict topic	0.010	0.005	2.067	0.042*
Time finishing tangram	-0.908	0.967	-0.939	0.350
Interpersonal mindfulness predicting dependent variables				
Affective aspect				
Positive Emotion	0.152	0.037	4.044	< 0.001***
Negative Emotion	-0.092	0.046	-2.020	0.047*
Stress	-0.110	0.033	-3.320	0.001**
Cognitive aspect				
Communication Satisfaction	0.162	0.023	6.939	< 0.001***
Empathy	0.441	0.050	8.669	< 0.001***
Psychological Distance	0.035	0.012	2.766	0.006**
Behavioral aspect				
Consensus in Conflict Topic	0.034	0.007	4.536	< 0.001***
Time Finishing Tangram	-3.328	1.580	-2.106	0.038*

Mindfulness was treated as a fixed effect across all models. Pre-test scores of positive emotion were included as a control variable in the post-positive emotion model, while pre-test scores of negative emotion were included as a control variable in the post-negative emotion model. Additionally, the intercept was treated as a random effect across all models. $N = 88$. * $p < .05$; ** $p < .01$; *** $p < .001$

Table 3 Actor effects of mindfulness on communication quality

Actor Effects	Estimated Coefficient (β)	SE	t	p
Intrapersonal Mindfulness				
YM _{intra} → YA Stress	-0.095	0.030	-3.182	0.002**
OM _{intra} → OA Positive Emotion	0.075	0.036	2.067	0.042*
OM _{intra} → OA Communication Satisfaction	0.081	0.029	2.765	0.007**
OM _{intra} → OA Empathy	0.204	0.079	2.584	0.012*
Interpersonal Mindfulness				
YM _{inter} → YA Communication Satisfaction	0.139	0.041	3.371	0.001**
YM _{inter} → YA Positive Emotion	0.176	0.066	2.667	0.009*
YM _{inter} → YA Empathy	0.382	0.079	4.855	< 0.001***
OM _{inter} → OA Communication Satisfaction	0.181	0.032	5.646	< 0.001***
OM _{inter} → OA Positive Emotion	0.177	0.047	3.775	< 0.001***
OM _{inter} → OA Negative Emotion	-0.253	0.062	-4.045	< 0.001***
OM _{inter} → OA Empathy	0.533	0.077	6.972	< 0.001***
OM _{inter} → OA Stress	-0.123	0.053	-2.301	0.024*
OM _{inter} → OA Consensus	0.037	0.010	3.557	< 0.001***
OM _{inter} → OA Time Finishing Tangram	-5.787	2.176	-2.660	0.009**

YM_{intra} = younger adult's intrapersonal mindfulness, OM_{intra} = older adult's intrapersonal mindfulness. YM_{inter} = younger adult's interpersonal mindfulness, OM_{inter} = older adult's interpersonal mindfulness. YA = younger adult, OA = older adult. Pre-test positive emotion scores are a control variable in the positive emotion model. For the sake of brevity, this table only presents significant Actor Effects. Other Actor Effects not shown did not reach statistical significance. $N = 88$. * $p < .05$; ** $p < .01$; *** $p < .001$

Table 4 Partner effects of mindfulness on communication quality

Partner Effects	Estimated Coefficient (β)	SE	t	p
Intrapersonal Mindfulness				
OM _{intra} → YA Stress	0.082	0.034	2.433	0.017*
OM _{intra} → YA Consensus in Conflict Topic	0.019	0.009	2.008	0.048*
Interpersonal Mindfulness				
YM _{inter} → OA Psychological Distance	0.051	0.020	2.340	0.021*
OM _{inter} → YA Consensus in Conflict Topic	0.027	0.012	2.305	0.024*

YM_{intra} = younger adult's intrapersonal mindfulness, OM_{intra} = older adult's intrapersonal mindfulness. YM_{inter} = younger adult's interpersonal mindfulness, OM_{inter} = older adult's interpersonal mindfulness. YA = younger adult, OA = older adult. Pre-test scores of psychological distance is included as a control variable in the positive emotion model. For the sake of brevity, this table only presents significant Partner Effects. Other Partner Effects not shown did not reach statistical significance. $N = 88$. * $p < .05$

communication satisfaction ($\beta = 0.081$, $SE = 0.029$, $t = 2.765$, $p = .007$), and greater empathy towards their communication partners ($\beta = 0.204$, $SE = 0.079$, $t = 2.584$, $p = .012$).

The results also showed that higher interpersonal mindfulness in younger adults was associated with higher communication satisfaction ($\beta = 0.139$, $SE = 0.041$, $t = 3.371$, $p = .001$), higher positive emotion ($\beta = 0.176$, $SE = 0.066$, $t = 2.667$, $p = .009$), and higher empathy ($\beta = 0.382$, $SE = 0.079$, $t = 4.855$, $p < .001$), while higher interpersonal mindfulness in older adults was associated with higher communication satisfaction ($\beta = 0.181$, $SE = 0.032$, $t = 5.646$, $p < .001$), higher positive emotion ($\beta = 0.177$, $SE = 0.047$, $t = 3.775$, $p < .001$), less negative emotion ($\beta = -0.253$, $SE = 0.062$, $t = -4.045$, $p < .001$), higher empathy ($\beta = 0.533$, $SE = 0.077$, $t = 6.972$, $p < .001$), lower stress ($\beta = -0.123$, $SE = 0.053$, $t = -2.301$, $p = .024$), higher consensus ($\beta = 0.037$, $SE = 0.010$, $t = 3.557$, $p < .001$), and less time to finish tangram tasks ($\beta = -5.787$, $SE = 2.176$, $t = -2.660$, $p = .009$).

The APIM model also controlled for intragenerational effects and examined whether the mindfulness of different generational communication partners would influence the communication quality, also known as *the partner effect* (see Table 4). The results indicated that higher intrapersonal mindfulness in older adults was associated with higher stress ($\beta = 0.082$, $SE = 0.034$, $t = 2.433$, $p = .017$) and higher consensus experienced by their younger partners ($\beta = 0.019$, $SE = 0.009$, $t = 2.008$, $p = .048$). In addition, higher interpersonal mindfulness in younger adults was associated with closer psychological distance perceived by their older partners ($\beta = 0.051$, $SE = 0.020$, $t = 2.340$, $p = .021$). Also, higher interpersonal mindfulness in older adults was associated with higher consensus experienced by their younger partners ($\beta = 0.027$, $SE = 0.012$, $t = 2.305$, $p = .024$).

Discussion

This study differentiated mindfulness into two sub-categories – intrapersonal mindfulness and interpersonal mindfulness – and investigated how they impact differently the three aspects (cognitive, affective, and behavior) of communication based on the Interpersonal Chemistry Model (Reis et al., 2021) between younger and older adults. We found that intrapersonal and interpersonal mindfulness improved intergenerational communication quality, answering our first research question affirmatively. Specifically, both intrapersonal and interpersonal mindfulness increased positive emotion (H1a), reduce stress (H1c), enhance communication satisfaction (H1 d), encourage more use of empathy (H1e), promote closer psychological distance (H1f) and lead to higher consensus on controversial topics (H1 g). Not only that, interpersonal mindfulness stands out and also decreased negative emotion (H1b) and sped up the completion of the cooperative tangram puzzle task.

At the affective level, both intrapersonal and interpersonal mindfulness increased positive emotion and decreased stress; however, lowered negative emotion after communication were only related to interpersonal mindfulness. These findings are similar to previous findings that intrapersonal mindfulness training can lead to more positive emotion and less stress and that the leader's interpersonal mindfulness positively correlates with the follower's satisfaction (Ramabramanian, 2017; Arendt et al., 2019). Additionally, this study's findings on the effects of negative emotion align with past research on mindfulness parenting, suggesting that interpersonal mindfulness can reduce psychological symptoms in parents, consistent with the results reported by Duncan et al. (2009).

From the cognitive perspective, higher communication satisfaction, higher empathy, and closer psychological distance were associated with both intrapersonal and interpersonal mindfulness. These findings are consistent with previous studies, suggesting a positive relationship between intrapersonal mindfulness and empathy (Cooper et al., 2020; Ardenghi et al., 2022; Bayot et al., 2020). This result of interpersonal mindfulness also aligns with Sofer (2018)'s view of the core mechanism of mindful communication, which posits that mindful communication is based on empathy, simultaneously empathizing with both one's own and others' inner worlds.

From the behavioral perspective, while intrapersonal and interpersonal mindfulness were associated with greater consensus on conflict issues, faster completion of cooperative tangram tasks was only related to interpersonal mindfulness. Previous studies, not differentiating intrapersonal versus interpersonal mindfulness, have suggested that mindfulness practices would effectively enhance cooperation between

individuals in diverse contexts, such as altruism and trust after a single mindfulness training session (Tappen, 2014), cooperative games (Doesum & Lange, 2016), negotiations (Masters-Waage et al., 2021), and inter-brain synchrony of interacting individuals during the cooperative tasks (Deng et al., 2023). Our finding is consistent with this wide range of research' outcomes and further points out that the interpersonal (but not intrapersonal) facet of mindfulness might be the key to facilitating the cooperation between the participants.

Another aim of this study was to compare whether there were differences in the enhancing aspects between interpersonal mindfulness and intrapersonal mindfulness on intergenerational communication quality. The results showed that interpersonal mindfulness affected more aspects of communication than intrapersonal mindfulness, with only interpersonal mindfulness associated with perceiving less negative emotion and faster speed in completing cooperative games. The possible reason is that interpersonal mindfulness is more sensitive to an interpersonal interaction situation than a solitary situation (Pratscher et al., 2018). Other studies have also found that interpersonal mindfulness improved interpersonal behaviors, including parenting behavior, parent-adolescent relationships, and friendship quality (Gannon et al., 2017; Bögels et al., 2014; Pratscher et al., 2018). Nevertheless, our data indicates a positive but medium correlation between intrapersonal and interpersonal mindfulness scores, similar to previous studies (Brown & Ryan, 2003), suggesting that these are related but distinct concepts. Thus, interpersonal mindfulness may reasonably capture more dimensions during interpersonal interactions than intrapersonal mindfulness, supporting our hypothesis 2.

When examining the impacts of intrapersonal and interpersonal mindfulness on communication quality as a whole, we also investigated whether they have varying effects on younger and older adults separately. The APIM found that within the dimension of intrapersonal mindfulness, younger adults with high intrapersonal mindfulness experience less stress. On the other hand, older adults with high intrapersonal mindfulness not only experience less stress but also have higher levels of positive emotion, greater communication satisfaction, and greater empathy towards their communication partners. Within the dimension of interpersonal mindfulness, younger adults with high interpersonal mindfulness experience higher levels of positive emotion, greater communication satisfaction, and greater empathy. Conversely, older adults with high interpersonal mindfulness experience higher communication satisfaction, higher positive emotion, and higher empathy but also experience lower stress, less negative emotion, higher consensus on conflict topics, and less time to finish the tangram task.

Overall, older adults with intrapersonal or interpersonal mindfulness displayed more improved facets of communication quality than younger adults, which supports our hypothesis 3. Previous studies suggested that mindfulness is more effective in older adults than young adults (Raes et al., 2015; Shook et al., 2017; Lenger et al., 2019). Examining the effects of intergenerational partners revealed that regardless of interpersonal or intrapersonal mindfulness, older adults' mindfulness facilitates younger adults' consensus on controversial topics. Surprisingly, Younger adults perceive more stress when facing older partners with higher intrapersonal mindfulness compared to those with lower intrapersonal mindfulness. One plausible explanation may be that, in traditional Chinese culture, older adults are highly respected for their wisdom and authority (Hsu, 1970). This interesting asymmetry suggests that, in Chinese culture, even with mindfulness, younger adults may still sense the feeling of pressure. However, younger adults' high level of intrapersonal mindfulness may offset the stress caused by older adults, causing less stress in these younger adults. Whether this result still holds true in other cultures awaits further investigation.

Implications

Ever since the innovative program called Mindfulness-Based Stress Reduction (MBSR), a structured eight-week course, was developed and first adopted for chronic pain (Kabat-Zinn, 1982, 1990), mindfulness-based programs and workshops are booming and have become a popular trend in world-famous companies (Gerszberg, 2023). In healthcare, MBSR increases employees' empathy and decreases their burnout (Moll et al., 2015). While these findings are impressive and encouraging, the classical approach focuses mainly on the intrapersonal dimension of mindfulness. However, when the nature of one's work involves frequent communication with others rather than working alone, focusing solely on intrapersonal mindfulness through MBSR may not be sufficient. Instead, incorporating elements of interpersonal mindfulness could lead to significant outcomes. This is supported by our findings, which indicate that compared to intrapersonal mindfulness, interpersonal mindfulness can predict communication quality across a broader spectrum, such as reducing negative emotion and improving collaboration efficiency.

In traditional family life, there exists a certain generation gap between different generations, which may lead to communication difficulties (Fong, 2007). The results of this study can be applied to shape better family communication, providing effective mindfulness intervention strategies for multigenerational families to enhance their intrapersonal mindfulness or interpersonal mindfulness, thereby

promoting the quality of intergenerational communication. For example, mindfulness practice can be cultivated as a habit among family members (such as focused attention mindfulness, mindful eating, etc.). Additionally, during family communication, certain interpersonal mindfulness rules can be established (such as being aware of whether one is criticizing others during communication, being conscious of and describing others' emotions, etc.) to reduce potential conflicts in communication.

Limitations and future directions

Our research aimed to address the intricate dynamics of family interactions and provide evidence-based guidance through a mindful approach to families experiencing intergenerational challenges and conflicts. However, due to practical considerations and the need to control for confounding variables—such as differing family roles (e.g., parent-child, husband-wife, grandparent-grandchild) and the varying durations of family conflicts—we opted for a more controlled approach by pairing strangers in an intergenerational communication dyad. Consequently, our findings may not fully capture the complexities of real-life family interactions. Nonetheless, our results can serve as a foundational step for future research to dig deeper into the nuanced and complex nature of family relationships.

In addition, the communication scenarios in our study were set within a laboratory situation. Whether the results of communicating under such a staged setting can be generalized fully to real-life circumstances, such as workplaces or family dynamics, requires further research with greater ecological validity. Also, after the communication session, participants self-reported their emotions, empathy, and attitude toward their partners and communication qualities. In the future, it will be valuable to add evaluation from the standpoint of the partners, such as perceived empathy.

Conclusion

Our findings suggest that both interpersonal and intrapersonal mindfulness can effectively enhance the quality of intergenerational communication, with a greater contribution from interpersonal mindfulness. Furthermore, either for interpersonal or intrapersonal mindfulness, it has been observed that older adults exhibit a greater number of areas of enhancement effect than their younger counterparts, and older adults aid in the establishment of consensus among younger adults regarding contentious issues. This study underscores the importance of mindfulness in intergenerational communication and recommends that future

communication tools be designed with mindfulness elements integrated into them.

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Data availability All de-identified raw data and code were stored in the OSF planform https://osf.io/df9rm/?view_only=f9411d8fa9fa4118b4897fbd013a98.

Declarations

Consent for publication Informed consent for publication was provided by the Research Ethics Committee at National Taiwan University (NTU-REC-202007HS058).

Ethics approval All experimental procedures and materials were approved by the Research Ethics Committee at National Taiwan University (NTU-REC-202007HS058).

Consent to participate All participants provided written informed consent and agreed to data publication terms before participating in the study.

Competing interests The authors have no conflicts of interest to disclose.

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